I Built a Full-Stack AI PDF Chatbot

A production-grade Retrieval-Augmented Generation (RAG) system that lets you upload any document and get accurate, grounded answers in real time.

Zero Hallucinations

Answers are strictly anchored to extracted document chunks via 1536-d vector search.

Real-Time Streaming

Token-by-token streaming responses powered by Vercel AI SDK and gpt-3.5-turbo.

Multi-Document Versatility

Tested across Resumes, Financial Invoices, Legal Contracts, and Industry Reports.

Swipe to see the Architecture & Live Document Demos ->

Full-Stack AI PDF Chatbot • Next.js 16 + LangChain + Pinecone

github.com/jayshreeganesh/pdf-chatbot

How It Works Under the Hood

From raw PDF binary to instant streaming response in 4 distinct stages:

01

Document Ingestion & Validation

Uploaded PDF binary is parsed page-by-page. Built-in validation catches empty or scanned image PDFs to prevent zero-text indexing.

02

Recursive Text Chunking

Splits content into 1,000-character chunks with 200-character overlap using RecursiveCharacterTextSplitter to preserve sentence semantics.

03

Vector Embeddings & Pinecone Indexing

Generates 1536-dimensional embeddings with OpenAI text-embedding-ada-002 and flushes previous document vectors to avoid data bleed.

04

Semantic Retrieval & Token Streaming

Top-5 cosine similarity search fetches the most relevant chunks. The LLM synthesizes the final response and streams tokens directly to the UI.

Screening Tech Candidates in Seconds

Mapped Document: resume_priya_sharma.pdf

Active Document: resume_priya_sharma.pdf

Status: 100% Indexed & Grounded

YOU (USER):

"What are Priya Sharma's top AI/ML skills and quantifiable achievements?"

AI CHATBOT (RETRIEVED FROM PDF):

- AI & ML Stack: PyTorch, HuggingFace Transformers, LangChain, Pinecone, vLLM, and Docker.
- Enterprise RAG Experience: Architected a vector knowledge base querying 500,000+ technical manuals.
- Quality & Reliability: Reduced hallucination rates by 34% by implementing Cohere Rerank.
- Fine-Tuning: Fine-tuned Llama-3 8B on proprietary legal datasets, achieving 91.4% accuracy.
- Education: M.Tech in Artificial Intelligence from the Indian Institute of Science (IISc), Bangalore.

BUSINESS VALUE: Reduces manual recruiter resume scanning from 8 minutes to 5 seconds per applicant.

Auditing Line Items & Wire Transfer Terms

Mapped Document: invoice_apex_digital_services.pdf

Active Document: invoice_apex_digital_services.pdf Status: 100% Indexed & Grounded

YOU (USER):

"Extract the invoice number, total amount due, line items, and payment instructions."

AI CHATBOT (RETRIEVED FROM PDF):

- Invoice Number: INV-2026-4401 | Payment Status: UNPAID | Due: November 15, 2026 (Net 30).
- Itemized Charges: NVIDIA H100 Cluster ($14,500), RAG Setup ($6,200), SLA Cloud Care ($3,800).
- Financial Breakdown: Subtotal: $27,000.00 | Tax (8.25%): $2,227.50 | Total Amount Due: $29,227.50.
- Bank Remittance: Silicon Tech Bank | Routing Code: STB1100234X | Account Number: 987654321098.

BUSINESS VALUE: Eliminates manual bookkeeping errors and prevents missed Net 30 vendor discount deadlines.

Extracting SLAs & Penalty Commitments

Mapped Document: contract_saas_service_agreement.pdf

Active Document: contract_saas_service_agreement.pdf

Status: 100% Indexed & Grounded

YOU (USER):

"What uptime is guaranteed in the SLA and what specific penalties apply for downtime?"

AI CHATBOT (RETRIEVED FROM PDF):

- Uptime Guarantee: Provider guarantees minimum monthly system uptime of 99.95%.
- Penalty Tier 1: If uptime drops below 99.5%, customer receives a 15% monthly billing credit.
- Penalty Tier 2: If uptime drops below 99.0%, customer receives a 30% monthly billing credit.
- Data Security Standard: Customer Data remains exclusive property; mandates SOC 2 and AES-256.
- Termination Requirement: Requires sixty (60) days prior written notice before contract expiration.

BUSINESS VALUE: Enables legal and procurement teams to review 40-page agreements in minutes instead of hours.

Synthesizing Complex Industry Trends

Mapped Document: report_global_renewable_energy_2026.pdf

Active Document: report_global_renewable_energy_2026.pdf

Status: 100% Indexed & Grounded

YOU (USER):

"What were the key quantitative findings for global solar and battery storage in 2026?"

AI CHATBOT (RETRIEVED FROM PDF):

- Record Growth: Added 510 Gigawatts (GW) globally in 2026, representing a 48% jump over 2024.
- Solar Dominance: Solar Photovoltaic (PV) accounted for 72% of all new renewable additions.
- Cost Reductions: Levelized Cost of Energy (LCOE) for utility-scale solar dropped to $0.028 per kWh.
- Battery Storage: Global Battery Energy Storage Systems (BESS) installed capacity crossed 180 GWh.

BUSINESS VALUE: Translates heavy market research whitepapers into executive bullet points and board-ready metrics.

Try It Out & Explore the Code

The complete project is open-sourced with test documents and setup instructions.

GitHub Repository:

https://github.com/jayshreeganesh/pdf-chatbot

What is included in the repository:

- Full Next.js 16 App Router code with real-time token streaming
- LangChain RAG pipeline with automatic Pinecone index cache flushing
- 6 pre-built test PDFs (Resumes, Invoices, Contracts, Research Reports)
- Complete USER_MANUAL.md with sample questions and prompt strategies

Like & Repost if you found this valuable!

Let me know in the comments: How is your team currently implementing RAG on documents?